

Pradhumn Vyas

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Work Experience

Applied AI Engineer Intern ATG/Banao Technologies

Oct 2025 - Feb 2026

- InterviewGod.ai Architecture & Proctoring:** Built multilingual AI interview platform utilizing Sarvam API for automated call screening, transcription, and LLM evaluation, reducing token consumption by **40%**; engineered Computer Vision pipelines to detect on-screen phones and analyze lip-sync dynamics, neutralizing deepfake attempts with **<2% false positives**.
- Vidya AI Voice Agent & RAG:** Engineered stateful, **sub-800ms** voice-to-voice latency conversational tutor integrating Gemini Live API and LangGraph; optimized WebSocket streams and deployed Pinecone vector DB for cross-session memory and high-speed context retrieval.
- Audio Intelligence & Sales Brain Pipeline:** Re-architected screen-scraping into a split-runtime serverless pipeline deploying Faster-Whisper (ASR) and Pyannote (diarization); engineered SpeechBrain ECAPA-TDNN voice biometrics with **>90%** confidence gating, achieving **0%** hallucinated speaker attributions. Offloaded ML inference to RunPod serverless GPUs, reducing active compute to 3–4 min/meeting; orchestrated Gemini 2.5 to extract Deal Health Scores, Failure Risk, and Seller Performance metrics into the database.
- Coding Assessment & Deployment:** Curated **5,000+** coding challenge dataset with Pinecone retrieval infrastructure ensuring hallucination-free automated evaluations; deployed microservices backend on AWS (ECS Fargate, ECR, Lambda) with GitHub Actions CI/CD, enabling **3x** peak traffic auto-scaling and cutting deployment cycles by **25%**.

Projects

Enterprise Customer Support AI Agent | LangGraph, RAG, Fast API, ChromaDB, Redis, Neo4j Apr 2026- May 2026

- Built a production-grade AI customer support agent using RAG architecture with LangChain and ChromaDB, processing PDFs, DOCX, and CSV into **1000**-token vector chunks achieving **85%+** retrieval precision and sub-2-second response time.
- Designed a 6-node multi-agent workflow using LangGraph with intent classification, sentiment analysis, Neo4j Graph RAG, and GPT-4o tool calling for real-time order status and refund lookups
- Containerized all **5** services (FastAPI backend, React frontend, ChromaDB, Neo4j, Redis) using Docker Compose and built a JWT-secured REST API with 5 endpoints handling chat, document upload, and ticket escalation
- Github Repository Link: <https://github.com/Pradhumn2004/Customer-Support-AI>

Personalized Health and Wellness Assistant | FastAPI, React/next.js, TensorFlow, Docker, SSE Mar 2026 - Apr 2026

- Built a full-stack personalized wellness platform with FastAPI backend and React/Next.js frontend, integrating 3+ REST APIs and nutrition databases delivering tailored diet and exercise recommendations to users across 5+ wellness categories.
- Implemented deep learning models using TensorFlow and PyTorch for activity recognition achieving **88%+** classification accuracy, with Scikit-learn pipelines processing **10,000+** data points for personalized wellness recommendations
- Automated CI/CD deployment pipeline using Docker and GitHub Actions reducing deployment time by **70%**, containerizing 4 services across development and production environments ensuring 99% uptime
- Github Repository Link: <https://github.com/Pradhumn2004/MuscleTab>

Social Media Post Automation Tool - Agentic AI | LangChain, Next.js, Celery, Redis, APScheduler Jan 2026 - Feb 2026

- Built an Agentic AI-powered social media automation tool using LangChain agents and Gemini 1.5 Pro API to autonomously generate, optimize, and schedule posts across 3+ platforms, reducing manual content creation time by **80%**
- Engineered an automated content pipeline fetching **50+** trending topics daily via Google RSS feeds, processing through Gemini 1.5 Pro for platform-specific post generation and scheduling via Celery + Redis task queue with APScheduler
- Developed Next.js 14 frontend with App Router and deployed containerized backend using Docker + GitHub Actions CI/CD, improving deployment efficiency by **35%** with reproducible builds and automated testing across 4 microservices
- Github Repository Link: <https://github.com/Pradhumn2004/drone-daily>

Achievements & Open Source

- Aden Hive (Core Contributor):** Engineered MCP tool integrations for the agent runtime harness, enabling secure API interactions with built-in state persistence.
- Bindu (Core Contributor):** Architected the DSPy adapter and expanded gRPC test coverage to facilitate secure Agent-to-Agent (A2A) communication via DIDs.

Technical Skills

Languages: Python, C++, SQL

AI & Machine Learning: PyTorch, TensorFlow, Hugging Face, LangChain, Scikit-Learn, NLP, Computer Vision, Transformer Architectures, Fine-tuning, Vector Databases, Embeddings, Inference Optimization

Backend & Distributed Systems: FastAPI, Celery, Redis, PostgreSQL, WebSockets, REST APIs

Cloud, DevOps & Observability: AWS(EKS, ECS, Lambda), Kubernetes, Docker, Prometheus, Grafana, Github Actions

Education

Lovely Professional University
Computer Science and Engineering — CGPA: 8.2
Mount Litera Zee School, 12th with Science
Greenvalley Public School, 10th with Science

2022 – 2026
Phagwara, Punjab
2021 – 2022 Bhilwara, Rajasthan
2019 – 2020 Bhilwara, Rajasthan